Ashmit Mukherjee

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Profile

Computer Science student at NYU Abu Dhabi building research platforms, machine-learning pipelines, and software systems. Experience spans human-AI experimental infrastructure, protein language model adaptation, multilingual NLP, and applied data analysis.

Education

New York University Abu Dhabi *Expected May 2027*
B.S. in Computer Science
Study Away: NYU Paris; NYU New York
Relevant Coursework: Natural Language Processing, Principles of Data Science, Statistics for Social and Behavioral Sciences, Data Structures and Algorithms, Computer Systems Architecture, Operating Systems

Research Experience

Research Assistant, Human-AI Collaboration, NYU Abu Dhabi *Feb 2026 to present*

- Lead the technical implementation of an online platform for a human-AI research project.
- Build, deploy, and maintain the system used to conduct the research and collect structured experimental data.
- Contribute to study design and translate research requirements into platform behavior, data workflows, and validation checks.

Research Assistant, eBRAIN Lab, NYU Abu Dhabi *Feb 2026 to present*

- Develop and evaluate parameter-efficient methods for adapting protein language models to protein-sequence tasks.
- Build training and evaluation pipelines and run multi-seed experiments on the NYU Abu Dhabi HPC cluster.
- Compare models and configurations to determine which results hold up across repeated runs.

Industry and Applied Research Experience

Machine Learning Engineer Intern, Zeek *Jun to Aug 2025*

- Built and evaluated customer-segmentation workflows using real-world data.
- Compared preprocessing and model choices, then used SHAP to inspect model behavior.
- Contributed to validation and evaluation procedures used in the model-development workflow.

Market Research Consultant, Global Media Alliance Broadcasting *Jun 2024*

- Led audience analysis, survey design, and customer-engagement analysis for a broadcasting company.
- Synthesized quantitative and qualitative findings into recommendations presented to senior leadership.

Selected Projects

Data Science Salary Prediction Platform *Fall 2025*

- Built a salary-prediction workflow on 742 cleaned Glassdoor job listings using PyCaret model comparison and XGBoost.
- Integrated SHAP feature-contribution analysis, MLflow experiment tracking, and a Streamlit interface for exploring model-based estimates.

Crime Data Analysis and Prediction Platform *Spring 2025*

- Modeled relationships between socioeconomic indicators and violent crime rates using data from 1,994 communities.
- Evaluated regression using a single holdout split and built a Streamlit application for descriptive exploration and model-based prediction; results represent observational associations.

Hinglish Named Entity Recognition Benchmark *Fall 2024*

- Contributed to a team project fine-tuning mBERT and XLM-RoBERTa for Hindi-English named-entity recognition on COMI-LINGUA.
- Worked on training, evaluation, and error analysis; encoder evaluation used 4,829 test examples, while zero-shot language-model baselines used a separate 100-example sample.

CAMP, Campus Asset Management Platform *Fall 2024*

- Contributed backend REST endpoints for staff workflows, equipment alerts, and dashboard statistics to a five-person campus asset-management project.
- Configured backend integration testing and client API utilities; served as Product Owner and Scrum Master in team sprint rotations.

Leadership and Campus Experience

First Year Program Facilitator, Office of Student Life, NYU Abu Dhabi *Aug to Dec 2024*

- Facilitated workshops, group discussions, peer activities, and weekly one-to-one check-ins for more than 20 first-year students.
- Connected students with academic, wellbeing, and campus resources and organized off-campus cultural excursions.

Front of House Staff, NYU Abu Dhabi Arts Center *Jan to May 2024*

- Managed guest check-in, crowd flow, and ticketing operations for events with more than 500 attendees.
- Supported VIP guests, performing artists, and last-minute operational changes.

Technical Skills

Programming: Python, C++, C#, JavaScript
Machine Learning and NLP: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, PyCaret, SHAP, LoRA, ESM-2
Data and Statistics: pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, cross-validation, regression modeling, experimental design
Systems and Tools: FastAPI, Empirica, REST APIs, MongoDB, Docker, Slurm, pytest, MLflow, GitHub Actions, Git, Streamlit, LaTeX